# Laboratory Report

Course: Discrete Control Systems
Lab No.: 3 – Design of Reference Signals
Discrete Generators
HDU ID: 22320605 | 22320630
Name: Cai Xuanhao | Zhu Chenhao
Variant: 22

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## 1. Design of command generator for sinusoidal reference signal

Given signal (Variant 22, Table 1):
$T = 0.25 \text{ s}, \quad A = 0.77, \quad \omega = 0.35 \text{ rad/s}$

$g(k) = 0.77 \sin(0.35 \cdot k \cdot 0.25) = 0.77 \sin(0.0875k)$

Let $\theta = \omega T = 0.0875$ rad
$\cos\theta \approx 0.99617, \quad 2\cos\theta \approx 1.99234$

State variables:
$\xi_1(k) = g(k)$
$\xi_2(k) = g(k+1)$

State-space model:

$$\begin{aligned} \xi_1(k+1) &= \xi_2(k) \\ \xi_2(k+1) &= -\xi_1(k) + 1.99234 \cdot \xi_2(k) \\ g(k) &= \xi_1(k) \end{aligned}$$

Matrix form:

$$\Gamma_d = \begin{bmatrix} 0 & 1 \\ -1 & 1.99234 \end{bmatrix}, \quad H = [1 \quad 0]$$

Initial conditions (phase = 0):
$g(0) = 0.77\sin(0) = 0$
$g(1) = 0.77\sin(0.0875) \approx 0.067375$

$$\xi(0) = \begin{bmatrix} 0 \\ 0.067375 \end{bmatrix}$$

## 2. Simulink scheme and result of command generator

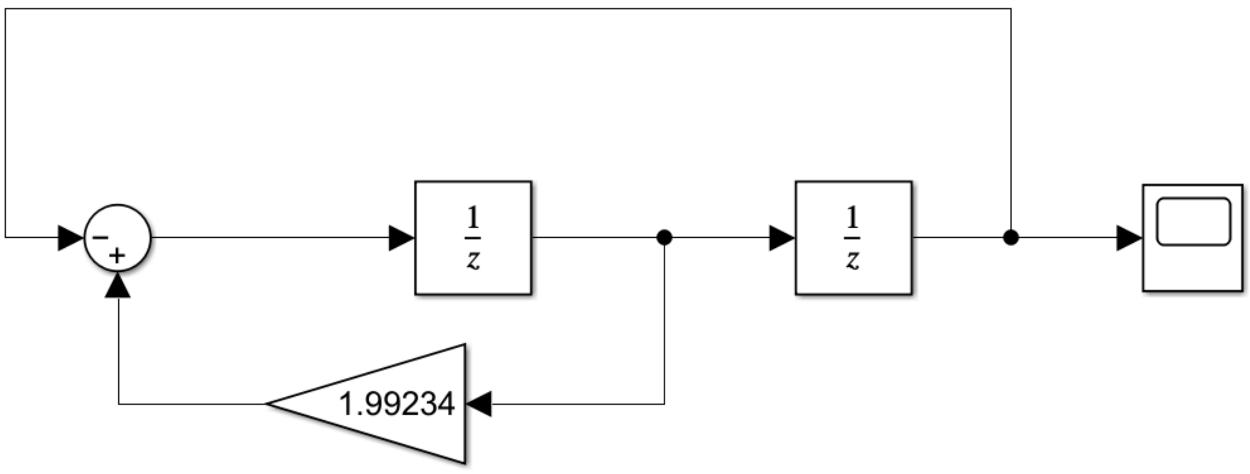


*Fig. 1. Simulink scheme of 2nd-order sinusoidal command generator

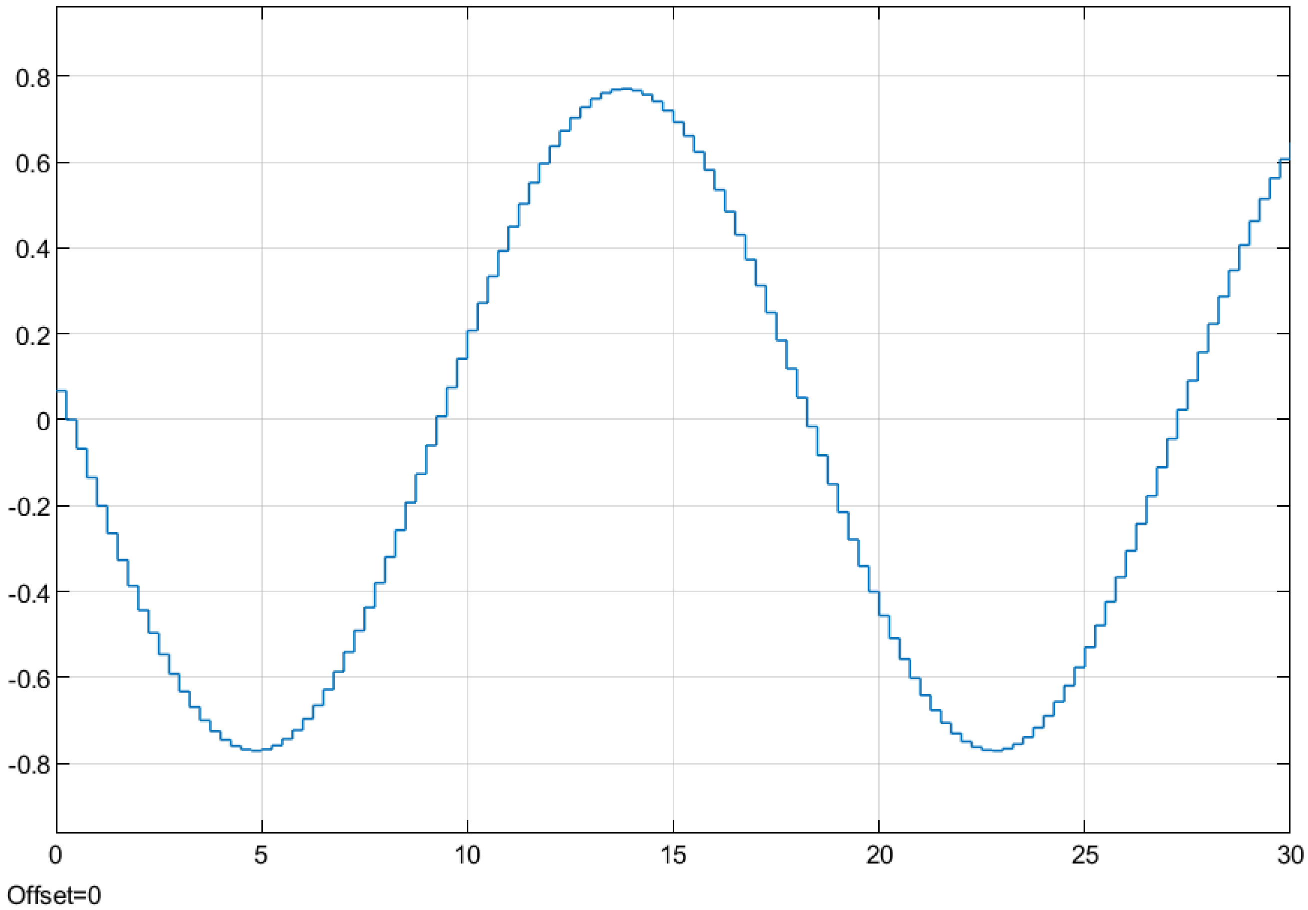


*Fig. 2. Generated reference signal g(k)

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## 3. Design an "input-state-output" discrete disturbance mathematical model using parameters listed in Table 2

Given disturbance (Variant 22):
$T = 0.25$ s
$d(k) = 0.8\sin(7kT) + 0.6\,kT = 0.8\sin(1.75k) + 0.15k$

The disturbance consists of a sinusoidal component and a linear ramp component.

### 3.1 Sinusoidal component $d_1(k) = 0.8\sin(1.75k)$

$$\begin{aligned} \theta &= 1.75 \text{ rad} \\ \cos\theta &= \cos(1.75) \approx -0.178246117 \\ 2\cos\theta &\approx -0.356492234 \end{aligned}$$

State-space model (2nd order):

$$\begin{aligned} x_1(k+1) &= x_2(k) \\ x_2(k+1) &= -x_1(k) - 0.356492\, x_2(k) \\ d_1(k) &= 0.8\, x_1(k) \end{aligned}$$

Initial conditions (zero phase):

$$\begin{aligned} &x_1(0) = 0.8\sin(0) = 0 \\ &x_2(0) = 0.8\sin(1.75) \approx 0.8 \times 0.983865 = 0.787092 \end{aligned}$$

### 3.2 Linear ramp component $d_2(k) = 0.15k$

State-space model (2nd order):

$$\begin{aligned} r_1(k+1) &= r_2(k) \\ r_2(k+1) &= r_2(k) + 0.15 \\ d_2(k) &= r_1(k) \end{aligned}$$

Initial conditions:

$$r_1(0) = 0, \quad r_2(0) = 0.15$$

### 3.3 Total disturbance

$$d(k) = d_1(k) + d_2(k)$$

## 4. Simulink scheme and result of disturbance generator

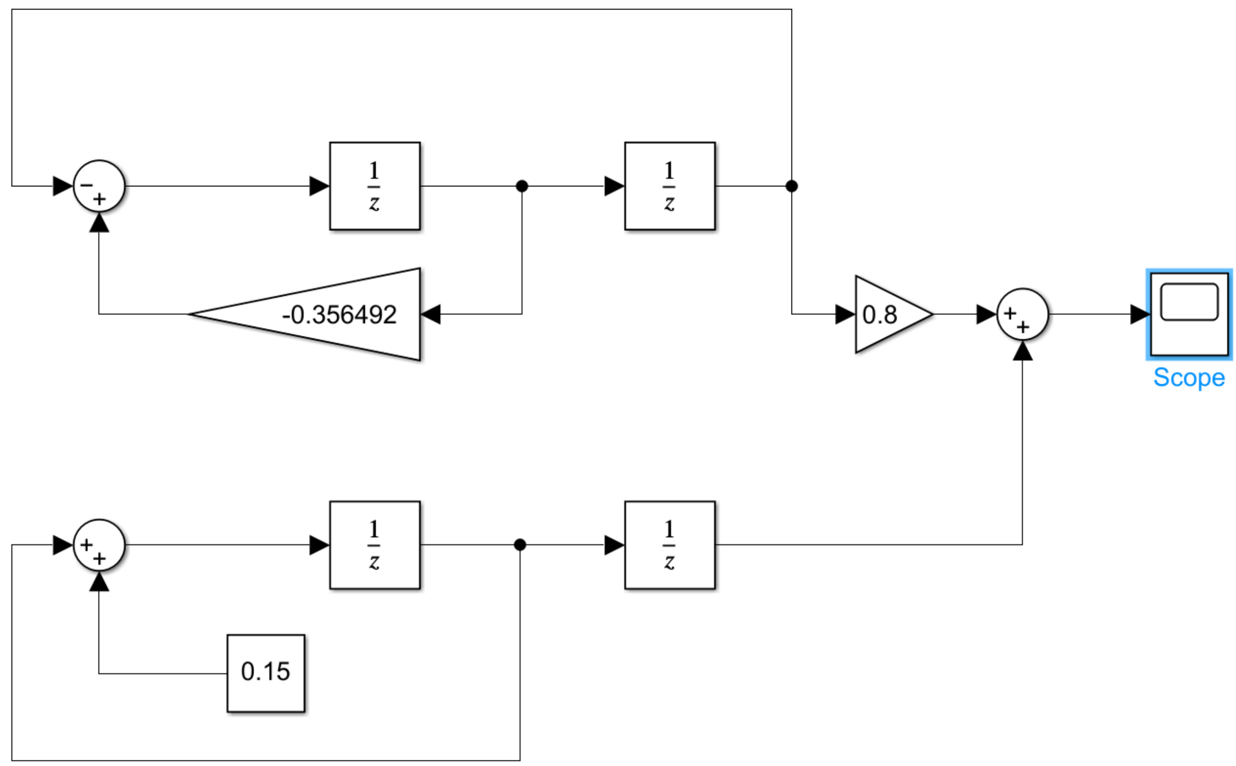


*Fig. 3. Simulink scheme of 4th-order disturbance generator (two 2nd-order subsystems in parallel)*

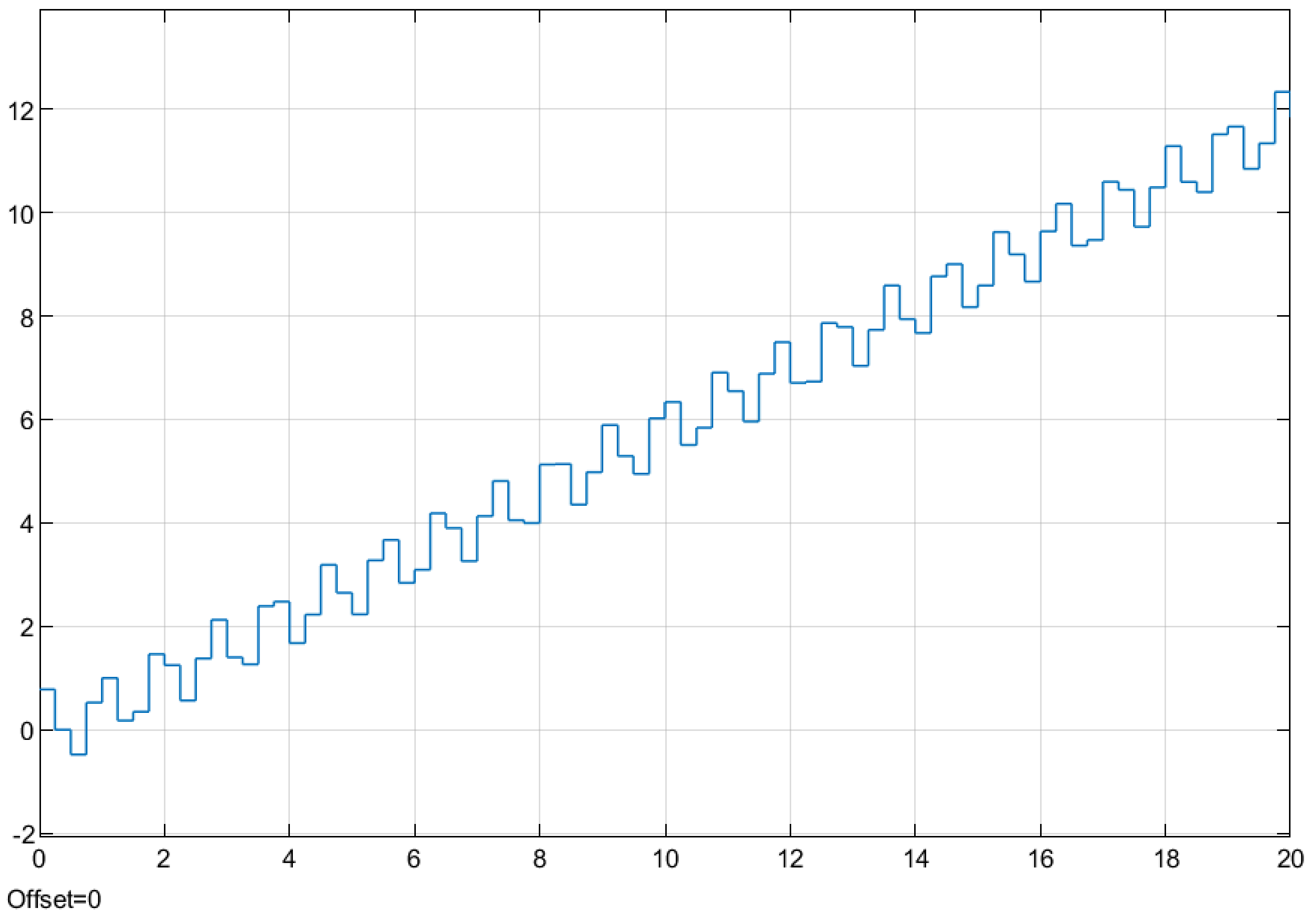


*Fig. 4. Generated disturbance signal d(k) — oscillating with linearly growing trend*

## 5. Conclusion

In this laboratory work, discrete state-space models were designed and implemented in Simulink for:

1. A sinusoidal reference signal $g(k) = 0.77\sin(0.0875k)$ using a 2nd-order generator.
2. A combined disturbance $d(k) = 0.8\sin(1.75k) + 0.15k$ using a 4th-order generator (sinusoidal + ramp).

Both generators were built using only Unit Delay, Gain, Sum, and Constant blocks. Simulation results fully match the theoretical signals: the command generator produces a stable discrete sine wave, and the disturbance generator correctly generates a sinusoid superimposed on a linear ramp. The work confirms that simple state-space structures are highly effective for generating typical reference and disturbance signals in digital control systems.

All simulations were performed with sampling period T = 0.25 s.